Assignment 3 & 4

(A) Write a script that takes a filename as input and displays information about the file, such as its size, permissions, owner, and modification time.

(B) Create a script that displays information about the system, such as the OS version, CPU, memory, disk usage, and network details.

(C) SHELL: You all have used shell (a shell is created when you launch terminal). So in this part, you shall write code for a very simple shell. Your shell

1. prints out a prompt of your choice (Some examples PathName>, PathName::, PathName>>>);

2. reads user input (expected commands of execution such as ls, rm, mkdir); These commands can have their parameters such as ls -l, ls -a, ls -al

3. parses the line into the program name and an array of parameters (separated by delimiters).

4. uses the fork() system call to spawn a new child process;

5. The child process then uses the exec() system call (or one of its variants) to launch the specified program;

6. The parent process (the shell) uses the wait() system call (or one of its variants) to wait for the child to terminate;

7. Once the child (the launched program) finishes, the shell repeats the loop by jumping to Step 1.

8. When user types exit, shell terminates.

(D). Write a C/C++ program to implement CPU scheduling. Data regarding process details need to be read from a file proc.dat

File format is as follows:

First line is number of processes N.

Each of next N lines contain ProcessID, Priority, Arrival time, followed by a sequence of CPU burst, IO burst. -1 indicates end of line. All these values are separated by comma. Following line indicates a process with id=2, priority 3, arrival time 4 and CPU bursts of 5, 4 and 7 interleaving with IO bursts of 2 and 3

2, 3, 4, 5, 2, 4, 3, 7, -1.

Last line contains value of quantum (by default and if this line is missing, quantum =1), Quantum is the interval at which timer interrupts.

Using this data, simulate following scheduling algorithms

a) FCFS
b) SJF (preemptive)
c) Priority (preemptive)

d) Round Robin

Your program should display progress of each process through a display with following symbols

C: running (CPU scheduled)

W: waiting for CPU

B; blocked for IO

R: running on IO.

For every process, history of its status has to be shown using above symbols.

(E). Write a program using C/C++/Java to simulate the FCFS, SJF (pre-emptive as well as non- pre-emptive approach). The scenario is: user may input n processes with respective CPU burst time and arrival time. System will ask the user to select the type of algorithm from the list mentioned above. System should display the waiting time for each process, average waiting time for the whole system, and final execution sequence.